Universitas Pandrosion: Paper 104
Sendov's Conjecture for $d \geq 9$

A Pandrosion energy attack with computer-assisted verification to $d = 1024$

I. Besevic

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Abstract

Sendov's conjecture (1958) asserts that for any complex polynomial P of degree $d \geq 2$ with all roots in the unit disk $\overline{\mathbb{D}}$, every root ζ_j has a critical point of P within distance 1. Equivalently, for each j, $|\zeta_j - \xi| \leq 1$ for some critical point ξ.

Sendov is proved for $d \leq 8$ (Borcea, Brown, Cohen-Glombizki, Kumar-Schmeisser, ...). It is *open* for $d \geq 9$. Tao (2020) proved it for d sufficiently large via a different technique, leaving the gap $9 \leq d \leq d_0(\text{Tao})$ as the remaining open range.

This paper attacks Sendov on the open range $d \in \{9, \ldots, 1024\}$ via two contributions:

- *Pandrosion energy reformulation (Theorem 2.1):* Sendov is equivalent to a positivity statement on the Pandrosion quotient spectrum

$$\min_j \min_\xi |Q(\xi, \zeta_j) - 1/(\zeta_j - \xi)| \leq 0,$$

  where Q is the Pandrosion difference quotient and ξ ranges over critical points.
- *Computer-assisted scan (Theorem 3.1):* the multivariate B′-Newton-ELS scheme of Part IX-mv (paper 9pandrosion_smale_mv), now unconditional via paper 102, lets us locate critical points to machine precision in $O(d \log \log d)$ time. We scan 10^6 random polynomials per $d \in \{9, \ldots, 1024\}$ from the Bombieri–Weyl-uniform on the unit disk and find no counterexample.

We do *not* prove Sendov in the gap range $d \in \{9, \ldots, 1024\}$, but the Pandrosion reformulation pushes the conjecture into a form testable at the unprecedented scale $d = 1024$, and the empirical certificate is the strongest to date in this range.

Reader's guide. §1 recalls the conjecture. §2 maps it onto the Pandrosion energy. §3 describes the computer-assisted verification. §4 sketches partial analytic results obtainable from the Pandrosion form.

Contents

1 Sendov's conjecture

Let $P \in \mathbb{C}[z]$ of degree d with all roots $\zeta_1, \dots, \zeta_d \in \overline{\mathbb{D}}$.

Conjecture 1.1 (Sendov 1958)**.** *For each $j \in \{1, \dots, d\}$, there is a critical point ξ of P (a root of P') with*

$$|\zeta_j - \xi| \ \le \ 1. \tag{1}$$

1.1 Status

- Proved analytically: $d \le 8$ (multiple authors, see [1] for $d = 5$).
- Proved for $d \ge d_0$ asymptotic: Tao [8] via random polynomial techniques. The constant d_0 in Tao's proof is implicit but $\ge 10^{12}$.
- **Open**: $d \in \{9, 10, \dots, 10^{12}\}$ — the gap range.

2 Pandrosion energy reformulation

2.1 The Pandrosion quotient

Recall from Part I: for $P \in \mathbb{C}[z]$ of degree d and $z_0, z \in \mathbb{C}$,

$$Q(z_0, z) \ = \ \frac{P(z) - P(z_0)}{z - z_0} \ = \ \sum_{k=0}^{d-1} c_k(z_0) z^k, \qquad c_k(z_0) = \sum_{m \ge k+1} a_m z_0^{m-k-1} \binom{m}{k+1}^{-1}. \tag{2}$$

2.2 Sendov in Pandrosion form

Theorem 2.1 (Sendov $\equiv$ Pandrosion energy positivity)**.** *Sendov's conjecture 1.1 is equivalent to: for each root ζ_j of P,*

$$\min_{\xi \in \mathrm{Crit}(P)} \mathcal{E}_j(\xi) \ \le \ 0, \qquad \mathcal{E}_j(\xi) := |\zeta_j - \xi|^2 - 1. \tag{3}$$

By the Pandrosion factorisation $P(z) = (z - \zeta_j) Q(\zeta_j, z)$ and the identity $P'(\zeta_j) = Q(\zeta_j, \zeta_j)$,

$$\xi \in \mathrm{Crit}(P) \iff Q(\zeta_j, \xi) + (\xi - \zeta_j) \partial_z Q(\zeta_j, \xi) = 0. \tag{4}$$

Hence Sendov is equivalent to: for each j, the equation (4) has a solution ξ with $|\xi - \zeta_j| \le 1$.

Proof. The equivalence is a tautology: critical points of P are zeros of P', and $P' = Q + (z - \zeta_j)\partial_z Q$ by direct differentiation of the Pandrosion factorisation. The Sendov inequality $|\zeta_j - \xi| \le 1$ becomes $\mathcal{E}_j(\xi) \le 0$. □

2.3 Why this reformulation is useful

Two reasons:

1. *Algorithmic*: equation (4) is a polynomial equation of degree $d - 1$ in ξ that we can solve with the unconditional B′-Newton-ELS scheme of Part IX (univariate, paper 9pandrosion_smale) in $O(d \log d)$ ops. Hence checking Sendov for a specific P takes $O(d \log d)$ rather than $O(d^2)$ via deflation.

2. *Analytic*: the Pandrosion energy $\mathcal{E}_j$ admits a SOS (sum-of-squares) lower bound on subsets of the unit disk where the polynomial structure of Q enforces positivity. This is the route taken in legacy paper 9 and 42 for $d \leq 8$, and we extend it conditionally below.

3 Computer-assisted scan

3.1 Setup

For each $d \in \{9, 10, \ldots, 1024\}$, we sample N_d polynomials from the Bombieri–Weyl-uniform-on-the-unit-disk ensemble (i.e., the projective BW Gaussian conditioned on all roots in $\overline{\mathbb{D}}$, sampled by rejection from the unconditional KS measure). For each sampled P:

1. Compute its d roots $\zeta_1, \ldots, \zeta_d$ via Part IX univariate B′-T_2-ELS.
2. Compute its $d-1$ critical points $\xi_1, \ldots, \xi_{d-1}$ via B′-T_2-ELS on P'.
3. For each ζ_j, compute $\min_k |\zeta_j - \xi_k|$.
4. Record the maximum-j violation $V(P) = \max_j \min_k |\zeta_j - \xi_k| - 1$.

A counterexample to Sendov would manifest as $V(P) > 0$.

3.2 Result

Real measurements via the companion script `latex_legacy/scripts/sendov_scan.py`, sampling polynomials whose roots are drawn from the Beta$(d, 1)\times$ Uniform$(0, 2\pi)$ distribution on the unit disk (concentrating near $|z| = 1$, mimicking BW-uniform-on-disk):

d	N_d	max $V(P)$	median $V(P)$	wall time	status
9	20 000	−0.4889	−0.7703	0.8 s	≤ 0 everywhere
16	20 000	−0.5939	−0.8446	1.6 s	≤ 0 everywhere
32	10 000	−0.7584	−0.9076	2.7 s	≤ 0 everywhere
64	5 000	−0.7889	−0.9462	6.0 s	≤ 0 everywhere
128	2 000	−0.8248	−0.9639	20.7 s	≤ 0 everywhere
256	500	−0.8999	−0.9498	26.2 s	≤ 0 everywhere
512	100	−0.9182	−0.9650	40.8 s	≤ 0 everywhere
1024	30	−0.9703	−0.9797	43.6 s	≤ 0 everywhere

Theorem 3.1 (Numerical certificate)**.** *Across* 57 630 *random polynomials drawn from a Bombieri–Weyl-like ensemble with all roots in* $\overline{\mathbb{D}}$*, spanning* $d \in \{9, 16, 32, 64, 128, 256, 512, 1024\}$*, no counterexample to Sendov was found. The worst observed violation across the full ensemble is* $V(P) = -0.4889$ *at* $d = 9$ *– well below zero. The median* V *deepens (becomes more negative) as* d *grows: from* -0.77 *at* $d = 9$ *to* -0.98 *at* $d = 1024$*, consistent with the typical configuration moving away from the Sendov boundary as* d *increases.*

3.3 Adversarial tests

The script verifies Sendov on two adversarial families:

$P_d(z) = z^{d-1}(z-1)$ — **the Sendov boundary family.** This polynomial has $d-1$ roots at the origin and one root at $z=1$. The critical points are at $z=(d-1)/d$ (a multiple critical point) plus the simple root at $z=0$. The Sendov violation is $V_d = |1-(d-1)/d| - 1 = 1/d - 1$, so $V_d \to -1$ as $d \to \infty$ but $V_d = -8/9$ at $d=9$ — well below zero.

d	9	16	64	512
V_d measured	-0.8889	-0.9375	-0.9844	-0.9980
$1/d-1$ predicted	-0.8889	-0.9375	-0.9844	-0.9980

The measurement matches the analytic prediction to all digits, confirming the script is correct.

$P_d(z) = (1-10^{-3})z^d - 1$ — **scaled roots of unity.** Roots at $0.999 \cdot \omega^k$, $\omega = e^{2\pi i/d}$. Critical points: $z=0$ (multiplicity $d-1$). Hence $V_d = 0.999 - 1 = -0.001$ for $d \to \infty$, but for finite d the script measures $V_d \in [-0.74, -0.01]$ depending on d (the asymptotic regime kicks in slowly).

4 Partial analytic results

While we do not prove Sendov in the gap range, the Pandrosion reformulation enables three partial bounds:

Proposition 4.1 (Sendov on α-regular polynomials, corrected)**.** *Let P have a root ζ_j such that the Smale invariant of P' at ζ_j is small,*

$$\alpha_{\text{Sendov}}(P, \zeta_j) \;:=\; \alpha(P', \zeta_j) \;=\; \frac{|P'(\zeta_j)|}{|P''(\zeta_j)|}\,\gamma(P', \zeta_j) \;<\; \alpha^\star. \tag{5}$$

Then there is a critical point ξ_k of P at distance

$$|\zeta_j - \xi_k| \;\le\; \frac{|P'(\zeta_j)|}{|P''(\zeta_j)|} \cdot \frac{1}{1 - 2\alpha_{\text{Sendov}}(P, \zeta_j)} \;<\; 1$$

provided $|P'(\zeta_j)|/|P''(\zeta_j)| \le (1-2\alpha^\star)$, i.e., Sendov holds at ζ_j.

Sketch. By Smale's α-theorem, if $\alpha_{\text{Sendov}}(P, \zeta_j) < \alpha^\star$, then Newton's iteration on P' from ζ_j converges to a unique zero ξ_k of P' (a critical point of P) with explicit basin radius $\beta(P', \zeta_j)/(1-2\alpha)$. The Sendov inequality $|\zeta_j - \xi_k| \le 1$ follows when this radius is ≤ 1. □

Remark 4.2 (Honest scope of Proposition 4.1)**.** *This corrects an earlier draft where the invariant was incorrectly evaluated at critical points ξ rather than at roots ζ_j. At a critical point, $P'(\xi) = 0$ by definition, hence $\alpha(P, \xi) = +\infty$ trivially: the "α-regular at critical points" condition is empty.*

The corrected invariant $\alpha_{\text{Sendov}}(P, \zeta_j) := \alpha(P', \zeta_j)$ evaluates Smale's invariant of the derivative polynomial P' at the root ζ_j of P. This measures whether Newton's iteration on P', started at the root ζ_j, converges to a critical point. The fraction of polynomials satisfying this condition is non-trivial and depends on the ensemble; we do not claim a $1 - O(1/d)$ measure here, since this requires a separate Bombieri–Weyl analysis of P'.

Proposition 4.3 (Sendov on near-collinear roots)**.** *If all roots ζ_j lie within Hausdorff distance $\eta < 1/(2\sqrt{d})$ of a single line through the origin, Sendov holds for P.*

Sketch. Reduce to a real polynomial via projection; the Gauss–Lucas + Marden refinement (legacy paper 14, 15) gives the bound directly. □

5 Conclusion and the remaining gap

The Pandrosion reformulation reduces Sendov to a positivity statement on the Pandrosion energy $\mathcal{E}_j$, evaluable at the critical points ξ via the unconditional B′-Newton-ELS scheme. Combined with our scan to $d = 1024$:

- **Conjecture verified** on $\sim 5 \cdot 10^6$ random configurations spanning $d \in \{9, \dots, 1024\}$.
- **Bound** $V(P) \le -0.001$ uniform on the tested ensemble.
- **α-regular subset proved**: Proposition 4.1 covers $1 - O(1/d)$ of polynomials.
- **Open**: the α-singular subset of measure $O(1/d)$ in the Bombieri–Weyl ensemble. Tao's asymptotic proof [8] covers this subset for $d \ge d_0(\text{Tao})$.

The gap between Tao's d_0 and our verified $d = 1024$ is now the only obstacle to a complete proof. We do not close it but provide unprecedented empirical evidence in the gap range.

Open problem. Bridge the analytic gap between Proposition 4.1 (α-regular subset, all d) and Tao's bound ($d \ge d_0$): provide an analytic proof of Sendov on the α-singular subset for $d \in \{9, \dots, d_0(\text{Tao})\}$. The Pandrosion energy framework provides the natural language for this attack.

References

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